

Assignment - 4 (Inheritance)

Que list 10 inheritance examples.

1) Vehicle System:

- Parent class : vehicle

attribute : Vehicle ID, brand, model, Fuel Type, color

methods : startengine(), stopengine(), accelerate(), brake().

- Child class (is-a)

A) car : Number of doors, trunk capacity, is sedan,

B) Truck : Payload capacity, towing capacity, No. of Axles.

C) motorcycle : has sidecar, handlebar type, engine displacement.

2) Employee management System

- Parent class : employee

attribute : employee ID, name, department, joining date

method : calculateSalary(), applyLeave(), updateProfile().

- Child class (is-a)

A) Full Time E : Annual Bonus, medical benefits, Pension Plan

B) Contract E : contract duration, hourly rate, agency name

C) Intern : stipend, university name, internship duration

3) electronic Devices

Parent class : electronic Device

attributes : deviceID, manufacturer, powerconsumption,
method : poweron(), poweroff(), resetdevice().

child class (is-a)

A) Smartphone : Screensize, camera resolution, OS

B) Laptop : Ramsize, Processortype, battery life

C) Television : display type, resolution, refresh rate.

4) Animal kingdom

Parent class : Animal

attributes : species, age, gender, habitat

method : eat(), sleep(), makesound()

child class (is-a)

A) mammal : furcolor, isdomesticated, gestation period

B) Bird : wingspan, canfly, breacktype

C) Fish : scaletype, watertype, ispredator.

5) Bank account

Parent class : Bank account

- account holder name : string, accountNo : int, C-Balance : double
+ deposit() , withdraw() , check balance()

child class (is-a)

- A) Saving Account = min balance, interest rate: double, cal interest
- B) Current Account = overdraft limit, business name, overdraft
- C) Loan Account = loan amount, emi amount, pay emi.

6) E-commerce product system

parent class : Product

attribute : product ID, name, string, price, stock quantity,
methods : add to cart, update price(), view details()

child classes (is-a)

- A) clothing : size, material, color variant
- B) electronics : model No, Voltage requirement, is rechargeable
- C) grocery : expiry date, is perishable, weight

7) Hospital management

Base class : person

attribute : person ID, fullName, address, phn No, emails
method : getContactInfo(), updateAddress()

Derived class (is-a)

- A) Doctor : specialization, itemNumber, consultation fee,
- B) Patient : blood group, medical history, insurance provider.
- C) Nurse : Shift Timing, Assigned ward, certification level.

8) Library management system

Base class : LibraryItem

attribute : itemID, title, publisher, publication year
method : checkout(), returnItem(), reserve

derived class (is-a)

- A) Book : Author, isbn, No. of Pages
- B) Magazine : Issue No., category, Frequency
- C) DVD : director, runtime, region code

9) Shape system

Base class : Shape

attribute : color, isFilled, borderThickness

method : calculateArea(), calculatePerimeter(), draw()

Derived class (is-a)

A) Circle : radius, diameter

B) Rectangle : length, width

C) Triangle : base, height, side length

10) university system

Base class : User Account

attribute : Username, Password, account status, login

method : login(), logout(), changePassword()

Derived class (is-a)

A) Student : enrollmentNumber, currentSemester, GPA

B) Faculty : employee code, designation, research area

C) Staff : role, office location, reporting manager